

Beyond GLMMs

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Complex data in the social and health sciences



Again our example...

PASSI research question

How is the probability of having at **least one chronic disease** associated with age, sex, education and economic difficulties?

The response is binary:

$$Y_i = \begin{cases} 1, & \text{respondent } i \text{ declares at least one chronic disease,} \\ 0, & \text{otherwise.} \end{cases}$$

and we have a vector of P covariates $\mathbf{x}_i \in \mathbb{R}^P$.

We consider the **region** as the clustering level.

Symbol	Meaning
$k = 1, \dots, K$	region
$i = 1, \dots, n_k$	respondent within region k
Y_{ik}	binary chronic disease indicator
$\mathbf{x}_{ik} \in \mathbb{R}^P$	individual and contextual covariates
$p = 1, \dots, P$	covariates
π_{ik}	$\Pr(Y_{ik} = 1 \mid \mathbf{x}_{ik})$

GLMMs limitations

Cluster-robust estimator

Generalized Estimating Equation

Resampling methods

To sum up

GLMMs limitations



Recap: Logistic GLMM

$$Y_{ik} \mid u_k \sim \text{Bernoulli}\{\pi_{ik}(u_k)\}, \quad \text{logit}\{\pi_{ik}(u_k)\} = \mathbf{x}_{ik}^\top \boldsymbol{\beta} + \mathbf{u}_k,$$

with

$$\mathbf{u}_k \stackrel{\text{iid}}{\sim} \mathcal{N}(\mathbf{0}, \tau^2).$$

What the random intercept does

It shifts the **conditional log-odds** for every respondent in the same region k and thereby induces **within-region association** after u_k is integrated out.

```
m_glmm <- lme4::glmer(chronic ~ age + sex + education + economic_difficulty +  
  (1 | region), family = binomial(), data = passi)
```

Recap: A GLMM answers a conditional question

Conditional on the regional random effect and holding all other covariates fixed, $\exp(\beta_r)$ is the odds ratio associated with a one-unit increase in x_r .

To obtain a population-average effect, we compute the **marginal** probability:

$$\Pr(Y_{ik} = 1 \mid \mathbf{x}_{ik}) = \int \text{expit}(\mathbf{x}_{ik}^\top \boldsymbol{\beta} + u) dF(u) \neq \text{expit}(\mathbf{x}_{ik}^\top \boldsymbol{\beta}).$$

First limitation is conceptual

If the scientific target is **population-average**, the conditional GLMM coefficient is not automatically the desired effect.¹

¹Neuhaus, J. M. et al. (1991). A comparison of cluster-specific and population-averaged approaches for analyzing correlated binary data. *International Statistical Review*.

Why use a GLMM at all?

A GLMM can simultaneously provide:

- **conditional effects** and region-specific predictions;
- **partial pooling of regional effects**: estimates from regions with less information are shrunk more strongly towards the overall mean;
- an estimate of $\tau^2 = \text{Var}(u_k)$ which quantifies **residual heterogeneity between regions** on the log-odds scale;
- **likelihood-based** estimation and model comparison.

The price of this flexibility

These advantages require specifying **more than the marginal mean**: we must decide how unobserved regional heterogeneity enters the model, specify its distribution, and approximate the resulting likelihood numerically.

Misspecification of a GLMM (fixed part)

It specifies how the outcome is associated with the observed covariates: $\mathbf{x}_{ik}^{\top} \boldsymbol{\beta}$.

It may be **misspecified** because of:

- omitted covariates;
- incorrect functional forms;
- omitted interactions;
- an inappropriate link function.

Main consequence

$\hat{\boldsymbol{\beta}} \not\rightarrow \boldsymbol{\beta}_0$: $\hat{\boldsymbol{\beta}}$ may **not consistently estimate** the intended conditional associations.

⚠ Increasing the sample size does not generally recover the intended conditional associations.

A short detour: multiverse analysis

"All models are wrong, but some are useful."

— George E. P. Box



Misspecification of the fixed part is not specific to GLMMs: it is a general problem in regression modelling. Some modelling choices are unavoidable.

Multiverse analysis²

Instead of relying on a single specification, we consider a set of **scientifically plausible models** and examine how the estimated effect changes across them. It is a structured sensitivity analysis to model specification.

²Steege et al. Increasing transparency through a multiverse analysis. *Perspect. Psychol. Sci.*, 2016.

A short detour: multiverse analysis

- **PIMA**³: simultaneous inference across multiple GLM specifications;
- **PIMAX**⁴: extension to multiple mixed-model specifications.

```
remotes::install_github("angeella/pima")
mod1 <- glm(Y ~ X1 + X2 + X3 + Z1 + Z2, data = D)
mod2 <- glm(Y ~ X1 + X2 + X3 + poly(Z1,2) + Z2, data = D)
mods <- list(mod1 = mod1, mod2 = mod2)
res <- pima(mods)
summary(res)
```

⚠ It does not identify the “correct” model, but assesses whether the conclusions are **robust across plausible specifications**.

³Girardi, P. et al. (2024). Post-selection inference in multiverse analysis (PIMA): An inferential framework based on the sign flipping score test. *Psychometrika*

⁴Vesely, A., & Andreella, A. (2026). Post-Selection Inference for Multiverse Analysis in Mixed-Effects Models (PIMAX). arXiv preprint arXiv:2607.03225.

Misspecification of a GLMM (random part)

Consider, for example, a random-intercept model

$$u_k \sim F(0, \tau^2).$$

The **random part** may be misspecified if

- the assumed **distribution** of u_k is inappropriate;
- relevant **random slopes or correlations** are omitted;
- **dependence within regions** is not captured by the specified random effects.

Why does this matter?

Misspecification can affect $\hat{\tau}^2$, $\hat{u}_k$, $\widehat{\text{Var}}(\hat{\beta})$ and, in some settings, also $\hat{\beta}$ itself.

➔ Correctly specifying the **mean model** is therefore not enough: the assumed dependence structure also matters.

Misspecification of a GLMM (random part)

Random effects can represent several sources of heterogeneity and dependence, but their structure must be chosen.

Ongoing debate⁵⁶⁷ $\Rightarrow$ Parsimonious vs maximal random-effects structures

⚠ There is no universally optimal structure: the choice should be guided by the **study design**, the scientific question and the information available in the data.

⁵Barr, D. J. et al. (2013). Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language*

⁶Bates, D., et al. (2015). Parsimonious mixed models. *arXiv preprint arXiv:1506.04967*.

⁷Matuschek, H. et al. (2017). Balancing Type I error and power in linear mixed models. *Journal of memory and language*, 94, 305-315.

Misspecification of a GLMM (random part)

A published example

A study was retracted after the authors found that their original mixed models had **omitted relevant by-subject random slopes**.

After fitting models that accounted for this variation, the previously reported interactions were no longer statistically significant.



Although anecdotal, this illustrates that a wrong random-effects structure may produce:

underestimated s.e $\Rightarrow$ inflated Type I error $\Rightarrow$ different scientific conclusions.

⚠ General guidelines are available, but choosing the random-effects structure requires both substantive reasoning and empirical diagnostics.

Random-effects specifications for PASSI

1. Regional random intercept:

$$\text{logit}(\pi_{ik}) = \mathbf{x}_{ik}^\top \boldsymbol{\beta} + u_{0k}, \quad u_{0k} \sim N(0, \tau_0^2)$$

Regions may have different baseline risks, but covariate associations are assumed to be the same across regions.

2. Regional random intercept and slope:

$$\text{logit}(\pi_{ik}) = \mathbf{x}_{ik}^\top \boldsymbol{\beta} + u_{0k} + u_{1k}x_{ik}, \quad \begin{pmatrix} u_{0k} \\ u_{1k} \end{pmatrix} \sim N(\mathbf{0}, \boldsymbol{\Sigma}_u).$$

Both baseline risk and the association with x_{ik} may vary across regions.

⚠ With only about **twenty regions**, estimating random-slope variances and intercept–slope correlations may be unstable. A random-intercept model is more parsimonious, but may be too restrictive if associations actually vary across regions.

GLMM computational issues

GLMMs rely on the marginal likelihood:

$$L(\beta, \tau^2) = \prod_{k=1}^K \int \underbrace{\prod_{i=1}^{n_k} \Pr(Y_{ik} = y_{ik} \mid \mathbf{x}_{ik}, u)}_{\text{conditional likelihood}} \underbrace{f(u; \tau^2)}_{\text{random-effects distribution}} du$$

For most GLMMs, this integral has no **closed-form solution** and must be approximated numerically.

A typical warning returned by lme4 is:

```
Warning message: Model failed to converge with max|grad| = 0.012 (tol = 0.002, component 1)
```

⚠ The algorithm stopped, but the estimated gradient suggests that it may not have reached the likelihood maximum.

The numerical approximation and maximization of the likelihood become more **challenging** in the presence of:

- complex or high-dimensional random-effects structures;
- weakly identified variance and correlation parameters;
- nearly singular random-effects covariance matrices;
- sparse or highly imbalanced data;
- complete or quasi-complete separation for binary outcomes;
- sensitivity to starting values or optimizer settings.

➔ **Inaccurate inference**, increased computational time and resource use.

Convergence is a separate issue

What convergence tells us

Convergence indicates that the numerical algorithm has found a solution according to its stopping criteria.

It does not guarantee that:

- the fixed part is correctly specified;
- regional heterogeneity is adequately represented;
- the assumed distribution of the random effects is appropriate;
- all relevant within-region dependence has been captured;
- standard errors and statistical tests are reliable.

A converged model can still produce biased or difficult-to-interpret coefficient estimates, distorted variance estimates, and unreliable standard errors.

What can we do in practice?

1. Check the data and model specification

Center and scale continuous covariates, check separation and consider whether the random-effects structure is supported by the data.

2. Try a different optimizer

```
fit_bobyqa <- update(fit, control = glmerControl(optimizer = "bobyqa", optCtrl = list(
  maxfun = 2e5)))
```

3. Compare several optimizers

```
fits <- lme4::allFit(fit)
summary(fits)
```

Recommended diagnostic

If different optimizers produce **very similar** coefficients, variance components and log-likelihoods, the warning may be a false positive.

Speed and numerical accuracy

For many lme4 models, bobyqa is among the fastest available optimizers. However, no optimizer is uniformly best.

➔ **Agreement across optimizers is more informative.**

In `glmer()`, numerical accuracy is also controlled by `nAGQ`:

- `nAGQ = 0`: faster, but less accurate approximation;
- `nAGQ = 1`: Laplace approximation;
- `nAGQ > 1`: potentially more accurate, but slower.

```
fit_laplace <- glmer(..., nAGQ = 1)
fit_agq <- glmer(..., nAGQ = 10)
```

⚠ Increasing `maxfun` gives the optimizer more evaluations; it does not improve the approximation of the integral.

Simulation: misspecifying the regional random effects

Data are generated from the logistic mixed model

$$\text{logit}\{\Pr(Y_{ik} = 1 \mid u_{0k}, u_{1k})\} = -1 + \beta_1 x_{ik} + u_{0k} + u_{1k} x_{ik},$$

where $u_{0k} \sim N(0, 0.7^2)$, $u_{1k} \sim N(0, \tau_1^2)$, $u_{0k} \perp u_{1k}$.

What happens to the Type I error when regional variation in x_{ik} is omitted?

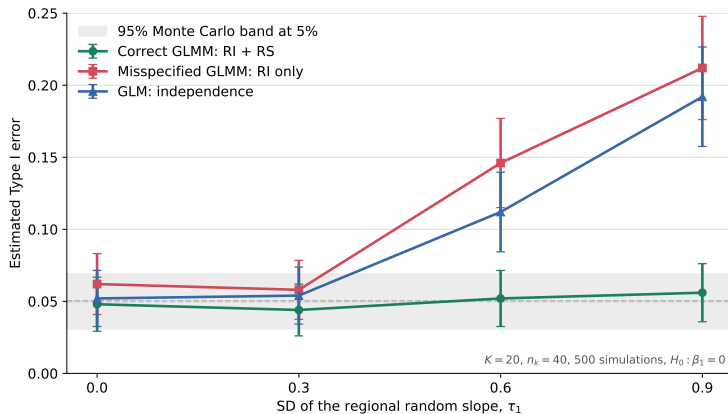
Simulation setting

- $K = 20$ regions;
- $n_k = 40$ respondents;
- $H_0 : \beta_1 = 0$;
- $\tau_1 \in \{0, 0.3, 0.6, 0.9\}$;
- 500 simulations.

Models fitted

1. correct GLMM: random intercept and slope;
2. misspecified GLMM: random intercept only;
3. GLM: regional dependence ignored.

Simulation: misspecifying the regional random effects



Across all simulation conditions, optimization warnings occurred in 36.4% of the correctly specified GLMMs and in 32.4% of the random-intercept-only GLMMs.

GLMM inference: what must be large?

Consider K independent regions, with n_k respondents in region k : $N = \sum_{k=1}^K n_k$.

Although GLMMs are fitted using all N observations, many inferential approximations rely on having a sufficiently large number of **independent clusters**:

$$K \rightarrow \infty \quad \text{not merely} \quad N \rightarrow \infty.$$

A large n_k helps

- estimation within each region;
- prediction of regional effects;
- precision of within-region associations.

But a small K remains, i.e., inference may be fragile for

- variance components;
- random-slope variability;
- region-level covariates;
- Wald and likelihood-ratio tests.

In PASSI, many respondents do not create more independent regions.

Another issue: separation in logistic GLMMs

Complete or quasi-complete separation occurs when a predictor, or a combination of predictors, almost perfectly predicts the binary outcome.

$$x_{ik} = 1 \implies Y_{ik} = 1 \quad \text{almost always.}$$

- Fixed-effect estimates may become extremely large.
- Standard errors and confidence intervals may become unreliable.
- Sparse outcome–predictor combinations within regions can aggravate the problem.
- Adding a random intercept does not guarantee that separation disappears.

⚠ Very large coefficients or standard errors, fitted probabilities close to zero or one, convergence warnings and a nearly singular Hessian.

Possible approaches

For ordinary logistic GLMs, bias reduction is available via `brglm2`. For GLMMs, penalized or Bayesian mixed models can be considered, with the regularization or

What if our target is marginal?

$$\Pr(Y_{ik} = 1 \mid \mathbf{x}_{ik})$$

Can we model the population-average association without explicitly specifying a random-effects distribution?

Spoiler: **YES**

- Cluster-robust covariance estimator
- Generalized estimating equations (GEE)

Cluster-robust estimator



CR0: the basic cluster sandwich

Starting from

$$\widehat{\text{Var}}(\hat{\beta}) = \hat{\mathbf{A}}^{-1} \hat{\mathbf{B}} \hat{\mathbf{A}}^{-1},$$

the simplest cluster-robust estimator uses

$$\hat{\mathbf{B}}_{\text{CR0}} = \sum_{k=1}^K \hat{\mathbf{U}}_k \hat{\mathbf{U}}_k^{\top}.$$

Intuition

CR0 treats the **regions**, rather than the individual respondents, as the **independent** units used to estimate variability.

⚠ CR0 is justified asymptotically as the number of independent regions K becomes large.

CR1: a simple finite-sample correction

With few clusters, CR0 tends to **underestimate sampling variability**. A simple solution is to inflate the entire CR0 covariance matrix:

$$\widehat{\text{Var}}_{\text{CR1}}(\hat{\beta}) = \frac{K}{K-1} \widehat{\text{Var}}_{\text{CR0}}(\hat{\beta})$$

Intuition

CR1 says:

“We only have a limited number of independent clusters,
so CR0 is probably somewhat too optimistic.”

It therefore applies the **same inflation to the whole covariance matrix**.

➔ Simple correction, but not distinguish between high- and low-leverage clusters.

Cluster HC3: correct more where leverage is high

Not all regions contribute equally to estimation. Some regions may have high leverage because of their covariate configuration, size, or design.

At the cluster level, HC3 uses a leverage adjustment of the working residuals:

$$\tilde{\mathbf{r}}_k^{\text{HC3}} \propto (\mathbf{I} - \mathbf{H}_{kk})^{-1} \hat{\mathbf{r}}_k,$$

where $\mathbf{H}_{kk}$ is the block of the hat matrix corresponding to region k . This produces leverage-adjusted cluster score contributions and, consequently, an adjusted meat:

$$\hat{\mathbf{B}}_{\text{HC3}} = \sum_{k=1}^K \tilde{\mathbf{U}}_k^{\text{HC3}} \tilde{\mathbf{U}}_k^{\text{HC3}\top}.$$

Intuition

High-leverage clusters can have residuals that are artificially too small. HC3 **inflates** their contribution to the estimated variance to compensate.

CR2: bias-reduced cluster-robust covariance

The basic cluster sandwich uses $\hat{\mathbf{U}}_k = \mathbf{U}_k(\hat{\beta})$, whose empirical variability can be underestimated in finite samples.

CR2 applies a cluster-specific adjustment. Schematically, $\tilde{\mathbf{U}}_k^{\text{CR2}} = \mathbf{C}_k \hat{\mathbf{U}}_k$.

The meat becomes

$$\hat{\mathbf{B}}_{\text{CR2}} = \sum_{k=1}^K \tilde{\mathbf{U}}_k^{\text{CR2}} \tilde{\mathbf{U}}_k^{\text{CR2}\top}.$$

The adjustment $\mathbf{C}_k$ depends on the design, leverage and a **working variance model**, and is calibrated to reduce finite-sample bias under that working model.

Intuition

Unlike CR1, CR2 does not apply the same correction everywhere: it uses the information in the design to correct each **cluster differently**.

What are CR1, HC3 and CR2 actually changing?

All methods start from the same sandwich structure:

$$\widehat{\text{Var}}(\hat{\beta}) = \hat{A}^{-1} \hat{B} \hat{A}^{-1}.$$

	What changes?	Intuition
CR0	-	Use empirical cluster-level variability
CR1	One global multiplier	Inflate CR0 because K is finite
HC3	Leverage-adjusted contributions	Correct more for influential clusters
CR2	Cluster-specific adjustments	Reduce finite-sample bias using the design and a working model

With few clusters, there are two problems

For cluster-robust inference, the relevant number of independent units is essentially K , not $N = \sum_k n_k$.

With **small** K , two things can go wrong:

1. **The covariance estimator can be biased.**

CR1, HC3, CR2

address this problem in different ways.

2. **The normal approximation can be too optimistic.**

Using

$$Z \approx \mathcal{N}(0, 1)$$

may still lead to excessive rejection.

Correcting the test as well as the covariance

Correcting the covariance is only one part of the problem.

Instead of $Z_r \approx \mathcal{N}(0, 1)$, we may use a **t reference distribution**.

For example, CR2 is commonly combined with

$$T_r = \frac{\hat{\beta}_r - \beta_{r,0}}{\widehat{\text{SE}}_{\text{CR2}}(\hat{\beta}_r)} \approx t_{\nu_r},$$

where ν_r is obtained using a **Satterthwaite approximation**.

Important

ν_r depends on the coefficient being tested and on the design. It can be substantially smaller than $K - 1$.



sandwich

- General framework for **sandwich covariance estimators**.
- Cluster-robust covariance through `sandwich::vcovCL()`.
- Allows HC-type corrections, including `type = "HC3"`



clubSandwich

- Designed for **cluster-robust variance estimation**.
- Cluster corrections through `clubSandwich::vcovCR()`
- Implements several corrections: `CR0`, `CR1`, `CR2`,...
- Provides small-sample tests, e.g. `clubSandwich::coef_test()`

PASSI example: cluster-robust covariance estimators in R

We keep the same marginal logistic model used before:

```
m_glm <- glm(chronic ~ age + sex, data = passi, family = binomial(link = "logit"))
```

and identify observations belonging to the same region.

```
# CR1
clubSandwich::coef_test(m_glm, vcov = "CR1", cluster = passi$region)

# CR2
clubSandwich::coef_test(m_glm, vcov = "CR2", cluster = passi$region)

# Cluster HC3
V_HC3 <- sandwich::vcovCL(m_glm, cluster = passi$region, type = "HC3", cadjust = FALSE)
lmtest::coeftest(m_glm, vcov. = V_HC3, df = length(unique(passi$region)) - 1)
```

➔ The estimated coefficients $\hat{\beta}_{\text{GLM}}$ do **not change**. Only $\widehat{\text{Var}}(\hat{\beta})$ and therefore standard errors, confidence intervals and tests change.

PASSI example: comparing standard errors

```
# CR1
V_CR1 <- clubSandwich::vcovCR(m_glm, cluster = passi$region, type = "CR1")
# CR2
V_CR2 <- clubSandwich::vcovCR(m_glm, cluster = passi$region, type = "CR2")
# Cluster HC3
V_HC3 <- sandwich::vcovCL(m_glm, cluster = passi$region, type = "HC3", cadjust = FALSE)

# Model-based GLM standard errors
sqrt(diag(stats::vcov(m_glm)))

# Cluster-robust standard errors
sqrt(diag(V_CR1))
sqrt(diag(V_HC3))
sqrt(diag(V_CR2))
```

For CR2, we can also directly obtain a small-sample corrected test:

```
clubSandwich::coef_test(m_glm,vcov = "CR2",cluster = passi$region,test = "Satterthwaite")
```

Which covariance correction should we use?

The choice depends mainly on the **number and structure of the clusters**.

In short^{8,9,10}

When K is limited, **CR2 + Satterthwaite** is a well-supported default. **HC3** provides a useful leverage-sensitive sensitivity analysis.

➔ There is no correction that is uniformly best in all designs.

⁸Bell, R. M. and McCaffrey, D. F. (2002). Bias reduction in standard errors for linear regression with multi-stage samples. *Survey Methodology*.

⁹Pustejovsky, J. E. and Tipton, E. (2018). Small-sample methods for cluster-robust variance estimation and hypothesis testing in fixed effects models. *Journal of Business & Economic Statistics*.

¹⁰Zeileis, A., Köll, S. and Graham, N. (2020). Various versatile variances: an object-oriented implementation of clustered covariances in R. *Journal of Statistical Software*.

What about mixed models?

LMMs

```
m_lmm <- lme4::lmer(y ~ x + (1 | cluster),
  data = my_data)

V_CR2 <- clubSandwich::vcovCR(m_lmm, cluster
  = my_data$cluster, type = "CR2")

clubSandwich::coef_test(m_lmm, vcov = "CR2",
  cluster = my_data$cluster, test = "
  Satterthwaite")
```

CR2 + small-sample inference is directly available.

GLMMs

```
m_glmm <- lme4::glmer(y ~ x + (1 | cluster),
  data = my_data, family = binomial)

library(merDeriv)

U <- sandwich::estfun(m_glmm) # cluster-wise
  scores

Br <- sandwich::bread(m_glmm) # bread
  component
```

⚠ not all GLMM random-effects structures are supported.

Take-home message

The **sandwich principle** is the same, but implementation becomes more complicated as the likelihood and random-effects structure become more complex.

Generalized Estimating Equation



GEE takes a different route

CR1, HC3 and CR2 start from the same working-independence GLM and modify only its estimated covariance.

GEE also uses the within-cluster association **when estimating** β .

For region k , let

$$\mathbf{Y}_k = (Y_{1k}, \dots, Y_{n_k, k})^\top.$$

We introduce a **working covariance** for

$$\text{Var}(\mathbf{Y}_k \mid \mathbf{X}_k) : \quad \boxed{\mathbf{V}_k = \mathbf{W}_k^{1/2} \mathbf{R}_k(\alpha) \mathbf{W}_k^{1/2}.}$$

For Bernoulli outcomes,

$$\mathbf{W}_k = \text{diag}\{\pi_{ik}(1 - \pi_{ik})\}.$$

Here $\mathbf{R}_k(\alpha)$ describes the assumed **within-region correlation**.

GEE: use the working correlation in estimation

GEE estimates β by solving

$$\sum_{k=1}^K \mathbf{D}_k^\top \mathbf{V}_k^{-1} (\mathbf{Y}_k - \boldsymbol{\mu}_k) = \mathbf{0}, \quad \text{where} \quad \mathbf{D}_k = \frac{\partial \boldsymbol{\mu}_k}{\partial \boldsymbol{\beta}^\top}.$$

Intuition

$\mathbf{V}_k^{-1}$ tells the estimating equation how much **independent information** different observations within a region are expected to provide.

- For the Bernoulli-logit mean model used here, if $\mathbf{R}_k = \mathbf{I}$, **GEE = GLM**.
- If observations are expected to be similarly correlated, we might use an **exchangeable** working correlation.

➔ Unlike CR1/HC3/CR2, changing $\mathbf{R}_k$ can also change $\hat{\beta}$.

What does the GEE working correlation look like?

Independence

$$\mathbf{R}_k = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

We pretend observations within the region are uncorrelated when estimating β .

Exchangeable

$$\mathbf{R}_k = \begin{pmatrix} 1 & \alpha & \alpha \\ \alpha & 1 & \alpha \\ \alpha & \alpha & 1 \end{pmatrix}.$$

Every pair within the region has the same working correlation α .

➔ **The working correlation need not be exactly correct:** if the marginal mean is correctly specified, robust sandwich inference can still be valid asymptotically as K grows.

PASSI example: GEE in R

```
library(geepack)

# observations from the same region must be contiguous
passi <- passi[order(passi$region), ]

m_gee <- geeglm(chronic ~ age + sex, id = region, data = passi, family = binomial(link = "logit"), corstr = "exchangeable", std.err = "san.se")

summary(m_gee)

# population-average odds ratios
exp(coef(m_gee))
```

Robust covariance or GEE?

	Cluster-robust	GEE
Mean model	marginal	marginal
How $\hat{\beta}$ is estimated	working independence	$R_k(\alpha)$
Does the dependence model affect $\hat{\beta}$?	No	Potentially yes
Main modelling choice	covariance correction: CR1, HC3, CR2, ...	working correlation: independence, exchangeable, ...

➔ Both approaches estimate **population-average effects**.

Robust covariance or GEE?

If there is **scientific knowledge** about the within-cluster association, a plausible working correlation GEE $R_k(\alpha)$ can be used to **improve efficiency** when estimating β . If the working correlation is misspecified, **robust sandwich inference for β** can still be asymptotically valid, under standard regularity conditions, if

the marginal mean is correct + clusters are independent + K is sufficiently large.

Avoid

Trying many working correlations or covariance corrections and reporting the one giving the most favorable p -value.

Good practice

Specify plausible choices scientifically and perform a **sensitivity analysis**: compare estimates, standard errors and substantive conclusions.

⚠ If neighboring regions remain spatially dependent, treating regions as independent clusters does not automatically solve the problem.

Resampling methods



Resampling methods

Approximate the **null distribution** of Z by repeatedly generating perturbed versions of the data or score contributions under H_0 . The perturbation is constructed while preserving the relevant dependence structure.

- **flip2sss**¹¹ approach for **GLMM** – **remmm** R package
- **wild cluster bootstrap**¹² for **LMM** – **fwildclusterboot** R package
- **CLIP**¹³ approach for **LMM** – **remmm** R package



¹¹Andreella, A., et al. (2025). Robust inference for generalized linear mixed models: a “two-stage summary statistics” approach based on score sign flipping. *Psychometrika*.

¹²MacKinnon, J. G. et al. (2023). Fast and reliable jackknife and bootstrap methods for cluster-robust inference. *Journal of Applied Econometrics*.

¹³Andreella, A., & Finos, L. (2026). Multivariate mixed models with model-free random effects. arXiv preprint arXiv:2604.27907.

To sum up



How should we proceed?

1. Start from the estimand.

- If cluster-specific heterogeneity or cluster-specific predictions are part of the target
→ **mixed model**.
- If the target is population-average and random effects are not themselves of interest
→ **marginal model**.

2. If the target is marginal, decide how to use the dependence.

- Do not model it in the point estimator → **GLM + cluster sandwich**.
- Use it for efficiency → **GEE + plausible working correlation**.

3. Count independent clusters, not observations.

- Large K : ordinary cluster sandwich may be adequate.
- Limited K : use a finite-sample covariance correction.

4. Check sensitivity. Compare conclusions across plausible robust procedures rather than relying on one asymptotic p -value.

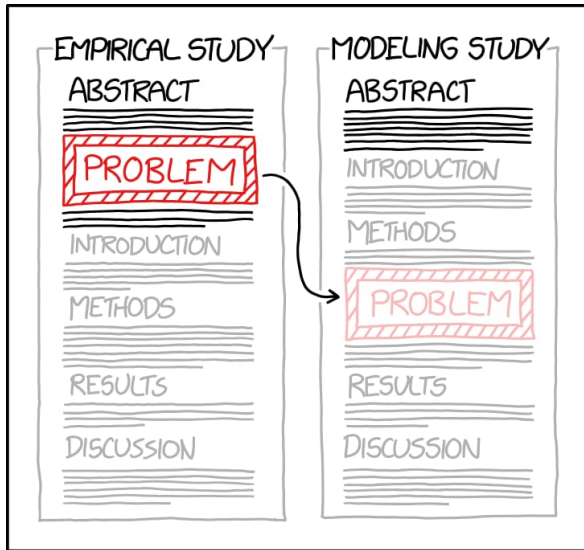
5. If model-based inference is fragile, consider resampling.

The central problem is **not** merely that observations are correlated.

We must decide **what effect we want to estimate** and **how much structure we are willing to assume** about the dependence.

Three ways forward

1. **Cluster sandwich:** keep the marginal GLM and correct its estimated uncertainty.
2. **GEE:** use a working correlation in estimation and a sandwich covariance for robust inference.
3. **Resampling:** approximate the null distribution without relying only on first-order asymptotic approximations.



A MATHEMATICAL MODEL IS A POWERFUL
TOOL FOR TAKING HARD PROBLEMS AND
MOVING THEM TO THE METHODS SECTION.

References: GLMMs and random-effects specification

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